Chapter

# 5

# GEOMETRY





## Learning objectives

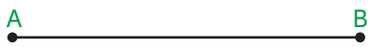
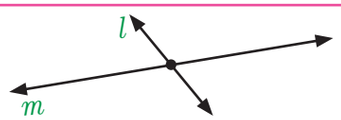
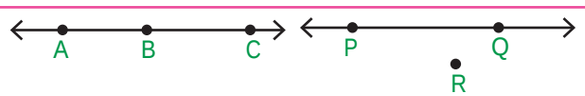
- To understand about adjacent angles, linear pair and vertically opposite angles.
- To understand transversal.
- To identify the different types of angles formed by a pair of lines with a transversal.
- To construct perpendicular bisector of a given line segment.
- To construct angle bisector of a given angle.
- To construct special angles such as 90°, 60°, 30° and 120° without using protractor.

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## Recap

## Lines

Let us recall the following concepts on lines and points which we have learnt in class VI.

| | |
|---|---|
| A **line** extends along both directions without any end. A line AB is denoted by $\overleftrightarrow{AB}$. Sometimes small letters like $l$, $m$, $n$ and so on are used to denote lines. |   |
| A **line segment** has two end points. The line segment 'AB' is represented by $\overline{AB}$. |   |
| A **Ray** is a line that starts from a point 'A' and extends without any end in a particular direction passing through 'B' which is denoted by $\overrightarrow{AB}$. |   |
| If two lines $m$ and $n$ are parallel, then we denote it as $m \parallel n$. **Parallel lines** never intersect each other. |   |
| When two lines have a common point they are called **intersecting lines** and that point is called the point of intersection of the given two lines. |   |
| If three or more points lie on the same line, then they are called **collinear points**; otherwise they are called **non-collinear points**. |   A,B and C are collinear points. P,Q and R are non-collinear points. |

### Try these

1. Complete the following statements.
   (i) A _____ is a straight path that goes on endlessly in two directions.
   (ii) A _____ is a line with two end points.



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